

Job Vacancy Management & Job Search System

Course: Web Programming

Group Size: 3 or 4 students

1. Assignment Overview

In this assignment, students will design and implement a **Job Vacancy Management and Search System**, focusing on the **employer side of the recruitment process**.

The system allows:

- **Employers** to create and manage **job vacancy postings**
- **Job seekers** to **search, filter, and view job vacancies**

This assignment builds directly on **Assignment 1 (Online CV Management & Search System)** and reuses:

- User roles
- Reference (lookup) tables
- Structured data principles

However, **CV creation and CV search are NOT part of this assignment**.

2. Learning Outcomes

Upon completion of this assignment, students will be able to:

- Design structured job vacancy data models
 - Implement dynamic job posting forms
 - Apply controlled vocabularies and lookup tables
 - Build multi-criteria job search functionality
 - Design employer-centric CRUD workflows
 - Optimize database schema for job searching
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3. System Roles

3.1 Employer

- Create and manage job vacancies

3.2 Job Seeker

- Search and view job vacancies (read-only)

3.3 Administrator

- Manage job postings and reference data
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4. Functional Requirements

4.1 Employer Features

4.1.1 Authentication

- Employers must register and log in securely
- Role-based access control is required

4.1.2 Job Vacancy Creation (Core Requirement)

Employers must be able to create **job vacancy postings** using a **structured form**. Each job vacancy must be stored in the database using **normalized and searchable fields**.

A. Basic Job Information

Field	Input Type
Job Title	Select
Job Category	Select
Employment Type	Select
Industry	Select
Job Level	Select (Junior / Mid / Senior)
Number of Openings	Number

B. Job Location (Structured – Mandatory)

Job location **must NOT** be stored as a single text field.

Field	Input Type
Country	Select

Field	Input Type
City / Province	Select
District	Select (optional)
Work Arrangement	Select (Onsite / Remote / Hybrid)

C. Salary & Benefits

Field	Input Type	Notes
Salary Range	Select	e.g., 1000–1500 USD
Salary Type	Select	Gross / Net
Benefits	Textarea	Free text allowed

D. Job Description (Free Text)

Employers must provide:

- Job responsibilities
- Required qualifications
- Preferred skills
- Additional notes

This is one of the **few free-text sections** allowed.

E. Required Skills (Mandatory)

- Employers must specify **required skills** for the job
- Skills must be selected from the **same predefined skill list used in Assignment 1**
- Employers may select **up to 5 required skills**

Field	Input Type
Skill	Select
Minimum Proficiency	Select

F. Education & Experience Requirements

Field	Input Type
Minimum Degree Level	Select
Minimum Years of Experience	Select

4.1.3 Job Vacancy Management

Employers must be able to:

- Create job postings
- Edit job postings
- Delete job postings
- Activate / deactivate job postings
- View list of their own job postings

Employers **must NOT** edit job postings created by other employers.

4.2 Job Seeker Features (Job Search & View Only)

Job seekers must be able to:

4.2.1 Search Job Vacancies by Multiple Criteria

- Keyword search (job title, description)
- Job category
- Location (country, city)
- Required skills
- Employment type
- Job level
- Salary range
- Work arrangement (onsite / remote / hybrid)

4.2.2 Combined Filters

- Multiple filters must be combinable using AND logic

4.2.3 Sorting Options

- Most recently posted jobs
- Salary (ascending / descending)
- Job title (alphabetical)

4.2.4 Job Vacancy Viewing

Each job vacancy view must display:

- Job title
- Employer information
- Location
- Salary range
- Required skills
- Job description
- Posting date

Job seekers **cannot apply** for jobs in this assignment.

4.3 Administrator Features

- Manage job vacancies

- Manage reference (lookup) tables:
 - Job categories
 - Job titles
 - Skills
 - Industries
 - Locations
 - Employment types
 - Job levels
 - Salary ranges
- Remove inappropriate or invalid job postings

5. Database & Data Requirements

- Use a **relational database**
- Job vacancy data must be **fully normalized**
- Reference tables must be used for all selectable fields
- Many-to-many relationship required between:
 - Job vacancies and required skills

✗ Storing job requirements or locations as plain text or JSON is **not allowed**.

6. Technical Requirements

- MVC architecture (or equivalent)
- Dynamic forms for:
 - Required skills
- Server-side validation
- Role-based access control
- Responsive UI
- Version control using Git

7. Deliverables

Item	Description
Source Code	Backend + frontend
Database Design	ER diagram + SQL schema
Technical Report	Job data model & design decisions
Demo Video	~5 minutes
Git Repository	Commit history required

8. Assessment Criteria

Category	Weight
Job vacancy creation & management	30%
Job search & filtering	25%
Database design & normalization	20%
Code structure & architecture	15%
UI & usability	5%
Documentation & demo	5%
Total	100%

9. Constraints & Notes

- No CV creation or CV search
- No job application workflow
- No matching or recommendation algorithm
- **Maximum 5 required skills per job**
- Employers only manage their own job postings

10. Academic Integrity

All submitted work must be original. External libraries or frameworks may be used but must be properly documented.